

LINE BREAKING AND PROCESS OPENING

Create an isolated, depressurized, drained, cleaned and verified boundary before opening piping, hoses, filters, covers or process equipment.

PURPOSE AND SCOPE

Create an isolated, depressurized, drained, cleaned and verified boundary before opening piping, hoses, filters, covers or process equipment.

KEY HAZARDS

 PRESSURE	 HOT PRODUCT	 CHEMICAL SPLASH	 FLAMMABLE VAPOUR
 TOXIC MATERIAL	 VACUUM	 TRAPPED VOLUME	 LINE OF FIRE

CRITICAL VALUES TO VERIFY IN THE FIELD

0 barg

Verified pressure before opening

0

Open drain or uncontrolled release

100%

Blind / isolation boundary verification



REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1910.119(f)(4), 1910.147 and the site line-break procedure require controlled process opening, hazardous-energy control, draining and verification.

OSHA 1910.119(f)(4)

OPENING PROCESS EQUIPMENT

Use a written, site-specific safe work practice for opening piping or equipment.

OSHA 1910.147

HAZARDOUS ENERGY

Isolate pressure, thermal, electrical, mechanical and potential energy, including reaccumulation.

SAHA LINE-BREAK PROSEDÜRÜ

DRAIN, PURGE AND BLIND

Engineering must define closed draining, flushing/purging and positive isolation for the product.

SDS / PROSES BİLGİSİ

PRODUCT AND PPE

Never assume residue is harmless; select barriers and PPE from SDS, temperature, pH, toxicity and flash risk.

MINIMUM EVIDENCE BEFORE START

- | | |
|---|---|
| ☐ Line-break permit and JSA | ☐ P&ID / isolation plan |
| ☐ Blind list | ☐ LOTO record |
| ☐ Drain / purge verification | ☐ Gas test / waste acceptance record |

SUPERVISOR DISCUSSION POINTS

- Where could pressure remain trapped?
- How did we verify zero energy by two methods?
- What is the stance and escape direction for first break?

POSITIVE ISOLATION, CLOSED DRAIN AND FIRST BREAK



P&ID - LOTO - Blind - Drain - Zero pressure - Shield

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. AVOID THE BREAK

Consider closed sampling, remote maintenance or alternatives that avoid opening equipment.

2. ESTABLISH POSITIVE ISOLATION

Create a reliable boundary using blinds/spades, spool removal or approved double-block-and-bleed; apply LOTO.

3. DRAIN, CLEAN AND VERIFY

Depressurize to a closed system, drain, purge/flush, and verify zero at gauge and an independent point.

4. MAKE A CONTROLLED FIRST BREAK

Loosen the first bolt away from the body under control; use shielding, drip containment and an escape route.

CRITICAL WARNING



A zero gauge can hide a blocked impulse line or trapped pressure. Never open a flange or cover without a second verification.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|--|---|
| ☐ Product / SDS known | ☐ P&ID and boundary verified |
| ☐ LOTO complete | ☐ Positive isolation installed |
| ☐ Closed drain ready | ☐ Purge / flush complete |
| ☐ Zero pressure double-verified | ☐ Shield / drip tray ready |
| ☐ Escape route clear | |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
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9			
10			
11			
12			